

System & Project Flow Architecture

The end-to-end execution flow for the **MERN Stack Mechanical & Robotics Portfolio** is organized into four core functional stages:

1. Visitor Experience & Data Retrieval Flow

1. **Client Initialization:** The browser loads the React single-page application (SPA), triggering the root layout, animated terminal hero component, and navigation bar.
- 2.
3. **Data Fetching:** On initial mount, React Query or Axios sends an asynchronous `GET /api/v1/projects` request to the Express API.
- 4.
5. **Cache & Render:** The backend retrieves project documents from MongoDB Atlas and returns a JSON payload containing CAD project data, technical tags, and media assets.
- 6.
7. **Interactive Filtering:** When a user selects a category (e.g., `Robotics`, `CAD/DFM`, `Automation`), the client-side state dynamically filters the Bento grid without requiring a full page refresh.
- 8.
9. **Modal Inspection:** Clicking a case study triggers a modal displaying CAD render carousels, assembly details, and kinematic descriptions.
- 10.

2. Contact & Lead Generation Flow

1. **Form Submission:** A visitor fills out the contact form (Name, Email, Subject, Message) and clicks submit.

- 2.
3. **Client-Side Validation:** Form state is verified for required fields and valid email syntax.
- 4.
5. **API Dispatch:** An asynchronous `POST` request is dispatched to `/api/v1/contact`.
- 6.
7. **Backend Security Layer:**
8.
 - **Rate Limiter:** `express-rate-limit` validates that the IP address has not exceeded the 5 requests / 15 minutes threshold.
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 - **Data Sanitization:** `express-validator` strips potential injection scripts and formats inputs.
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9. **Database Storage:** The sanitized inquiry is saved to MongoDB under the `Inquiry` collection with an initial status of `UNREAD`.
- 10.
11. **Notification Trigger:** A Nodemailer or SMTP service automatically dispatches a formatted email alert directly to `amruthvarunpyla@gmail.com`.
- 12.
13. **Client Feedback:** The user interface displays a success toast notification and resets the input fields.
- 14.

3. End-to-End Development Execution Flow

Phase	Milestone	Core Deliverables
Stage 1: Backend Setup	API & Data Layer	Initialize Node/Express server, configure MongoDB Atlas connection, write Mongoose models, and create database seed scripts with resume project data.
Stage 2: API Endpoints	REST Service & Security	Implement CRUD routes for projects, <code>/api/v1/contact</code> inquiry route, rate limiting, CORS configuration, and Nodemailer integration.

Stage 3: Frontend Foundation	UI Layout & Theme	Set up Vite + React with Tailwind CSS, configure font hierarchy (<i>Syne</i> , <i>Inter</i> , <i>JetBrains Mono</i>), and assemble responsive layout frames.
Stage 4: Interactive Components	Data Integration	Build Hero terminal typewriter, dynamically fetch and render project cards, and wire experience timelines.
Stage 5: Deployment & CI/CD	Production Release	Deploy the React client to Vercel/Netlify, deploy the Express API to Render/Railway, set up environment variables (<code>MONGO_URI</code> , <code>SMTP_KEYS</code>), and verify production CORS.

4. Directory & Code Organization Flow

Plaintext